KHAHP — Full Technical Evidence & Engineering Register

Public Consolidated Release v4.0 · 18 October 2026
Elite Cybersecurity Specialist · Malaysia
200+ hours of engineering work (owner-reported) · 44 sections · 308 families · 924 evidence gates

> **Current milestone:** Sections 01–40 are the owner-reported engineering workstream; Sections 41–44 are the remaining planned assurance and commercialization workstreams. The original gate sheet still records **0/924 formally signed A/B/C gates**. This is not a claim that all applicable acceptance evidence is complete. Section 39 BIOS is owner-completed and excluded from further retests.

1. Executive research and integrity statement

The project evaluates a defensively hardened Kicksecure/Debian 13 laptop, with daily-use USER and privileged SYSMAINT maintenance roles kept distinct. It aims to verify source provenance, real enforcement and persistence/recovery in the intended threat model. It does not imply stock Kicksecure has 924 vulnerabilities, that an audit run fixes controls, that a command returning zero is necessarily an effective-negative-test PASS, or that current development images are commercially certified.

Program measure	Number	Interpretation
Domains / sections	44	Seven control families per section
Control families	308	01.01 to 44.07
Source / runtime / persistence gates	924	A, B, C for each family
Historical mixed backlog	69	Review, manual, exception and change items

Program measure	Number	Interpretation
Tracked 01–38 historical action records	63	Neither a vulnerability count nor an automatic patch queue
Already supported configuration change events	At least 3	See Section 3; not total discovered vulnerabilities
Current formal signed gate total in provided register	0	Signature ledger has not been updated with newer tests
Commercial-image acceptance tests	60	Proposed future release tests, not executed

Boundary: The project owner reports more than 200 hours and considers Section 39 BIOS complete. This is an owner statement rather than an independently audited work-hour or firmware assertion. The 01–40 progress headline reports engineering scope, not universal A/B/C acceptance; Section 40 auditd/AIDE end-to-end testing has not been evidenced in the available reports.

2. Evidence rules and status vocabulary

Gate	Minimum evidence	Failure to prove does not imply
A — provenance	Package / configuration owner, original and changed value, intended policy, approved diff	No security feature exists
B — effective enforcement	Exact role, real-world allowed behavior, bounded denied behavior and observed outcomes	A test was passed merely because a unit was active
C — persistence and recovery	Relevant reboot, reconnect, update or isolated recovery plus rollback where required	A freshly observed setting will persist indefinitely

Statuses used in this public release: `OBSERVED`, `TEST-PASS`, `OWNER-COMPLETE`, `FIXED-IMMEDIATE-VERIFIED`, `REVIEW`, `MANUAL-REQUIRED`, `EXCEPTION`, `NOT-TESTED`, and `FORMALLY-ACCEPTED`. A source finding may be sound without being formally accepted. Do not treat `REVIEW` as `CONFIRMED-VULNERABILITY`. Formal acceptance requires an authorized reviewer, date and linked A/B/C proof.

The supplied CSV contained 924 `OPEN` gates. This edition **preserves that historical signature state** rather than counterfeiting signatures from command output. The private updated master supplies newer evidence candidates for owner/reviewer reconciliation.

3. Configuration changes, corrective actions and durable results

ID	Section	Change and observed result	Remaining boundary
CHG-07-01	07	`/tmp`: `nodev,nosuid` applied; prior USER reboot persistence observed	`noexec` intentionally excluded for compatibility; other mounts require own decisions
CHG-37-01	37	GNOME thumbnail generation disabled (`false` → `true`); later synthetic/reboot and PCManFM-Qt checks observed	Other applications, deletion policy and format-specific viewers require separate review
KHAHP-20261018-FSTAB-ORDER	07	`/boot` entry moved before `/boot/efi`; root `findmnt --verify --verbose`: no errors/warnings; backup retained	Validate after next operator-controlled reboot; change is not evidence of previous exploit

Not changed on purpose

- No blanket `/tmp noexec`; application and installer compatibility are explicit considerations.
- No sudden USER privilege escalation or weakening of SYSMAINT separation for testing convenience.
- No unreviewed global firewall/DNS policy cutover, Tor stop, network manager teardown or blanket USBGuard deny-all.
- No AIDE database reinitialization or invasive camera/microphone capture.
- Section 39 firmware/BIOS policy remains **owner-completed and excluded from retesting**.

4. High-value verification observations and limitations

Domain	Later dated technical evidence	Interpretation
OS/platform	Debian 13 trixie; Linux 6.12.111+deb13-amd64; zero failed systemd units in collected runs	Observed role- and time-scoped baseline
Kernel	`kptr_restrict=2`, `dmesg_restrict=1`, `ptrace_scope=3`, unprivileged BPF restricted, ASLR and protected file knobs observed	Effective values for inspected controls, not all kernel options proven
Secure Boot / modules	Secure Boot enabled in SYSMAINT snapshot; signed Tirdad build provenance observed	Signed-module rejection and future update behavior not yet signed off
Network resolver	Unbound listening on loopback, active, enabled and `unbound-checkconf` clean	Local resolver availability does not rule out app-specific DNS bypass
DNSSEC	Positive AD response and broken-DNSSEC SERVFAIL confirmed	DNS answer authenticity in bounded test; no transport confidentiality inference

Domain	Later dated technical evidence	Interpretation
DNS-over-TLS	Upstream Quad9 forwarding configuration, authenticated-name and TLS port 853 observations previously recorded	Negative certificate-path validation and all egress routes require independent test
nftables	Active rules; input/forward default-drop; OUTPUT policy accepts with explicit external direct DNS drops	Rules present, external UDP/TCP53 packet verdicts not fully signed
Tor	Tor active; SOCKS loopback listener present	Which apps are Tor-scoped and fail-closed not established by service activity
USBGuard	Device-specific authorized mouse and approved recovery/storage; no destructive policy changes	Keyboards/mice/storage allowed by owner policy; actual input and composite behavior require physical proof
Camera / mic	USER open() denied for two video nodes and one microphone capture node; no recording	Current nonprivileged direct-node test only, no guarantee against all privileged/ hardware paths
Bluetooth	rfkill soft blocks on both adapters and inactive service	Current block state observed
Browser	Firefox 158 active; `network.trr.mode=5`, WebRTC/prefetch restrictions and seccomp-filtered subprocesses observed	`/etc/apparmor.d/firefox` intentionally unconfined; no strict MAC claim
Desktop privacy	Thumbnail disabling retained; synthetic test and reboot persistence recorded; `swaylock` lock/unlock worked	Five-minute and lid/resume tests not passed by short lock test
Audit/AIDE	`dailyaidecheck.timer` enabled/active. SYSMAINT `auditctl -s`: audit enabled, PID 0; `auditctl -l`: no rules	Do not declare full audit-event or integrity alert lifecycle accepted

5. Implementation review by all 44 sections

Each section below states the **strongest supported later observation**, its limitation, and all seven control families with their exact A/B/C IDs. These compact tables replace the v3.0 boilerplate. The listed A/B/C identifiers are **acceptance requirements**, not assertions of signature or pass status.

01 — Platform / OS / Hardware Baseline

Observed implementation/evidence: Debian 13 (trixie), kernel 6.12.111+deb13-amd64 and zero failed systemd units were observed in the October 18 USER/SYSMAINT snapshots.

Acceptance boundary / next proof: Reconcile full hardware inventory and post-upgrade provenance to formal A/B/C acceptance.

Family	Exact technical scope	A/B/C evidence IDs
01.01	Operating-system identity and Debian/Kicksecure release baseline	KHAHP-01.01A · KHAHP-01.01B · KHAHP-01.01C
01.02	Kernel release, architecture and build identity	KHAHP-01.02A · KHAHP-01.02B · KHAHP-01.02C

Family	Exact technical scope	A/B/C evidence IDs
01.03	CPU model, topology and virtualization capability	KHAHP-01.03A · KHAHP-01.03B · KHAHP-01.03C
01.04	Physical memory and swap capacity baseline	KHAHP-01.04A · KHAHP-01.04B · KHAHP-01.04C
01.05	Storage controller, disk and filesystem topology	KHAHP-01.05A · KHAHP-01.05B · KHAHP-01.05C
01.06	UEFI/bare-metal/virtualization platform state	KHAHP-01.06A · KHAHP-01.06B · KHAHP-01.06C
01.07	Failed-unit, uptime, boot-time and baseline health state	KHAHP-01.07A · KHAHP-01.07B · KHAHP-01.07C

02 — Kicksecure Package Provenance

Observed implementation/evidence: The latest `dpkg --audit` returned no pending package-database problems. Provenance must be derived from real owning packages, not package-name guesses.

Acceptance boundary / next proof: Attribute outstanding permission-hardener policy and package-origin discrepancies.

Historical action references: `U01-003`.

Family	Exact technical scope	A/B/C evidence IDs
02.01	Core Kicksecure package installation and exact versions	KHAHP-02.01A · KHAHP-02.01B · KHAHP-02.01C
02.02	Supporting security-package installation and exact versions	KHAHP-02.02A · KHAHP-02.02B · KHAHP-02.02C
02.03	APT source files and repository configuration	KHAHP-02.03A · KHAHP-02.03B · KHAHP-02.03C
02.04	APT policy and repository origin attribution	KHAHP-02.04A · KHAHP-02.04B · KHAHP-02.04C
02.05	Package ownership of security-critical configuration files	KHAHP-02.05A · KHAHP-02.05B · KHAHP-02.05C
02.06	Package holds, partial installs and dpkg database health	KHAHP-02.06A · KHAHP-02.06B · KHAHP-02.06C
02.07	Local modifications versus package-owned baseline	KHAHP-02.07A · KHAHP-02.07B · KHAHP-02.07C

03 — USER / SYSMAINT Privilege Architecture

Observed implementation/evidence: The intended USER/SYSMAINT privilege separation was retained during testing; privileged maintenance ran as UID 0 in a separate maintenance context.

Acceptance boundary / next proof: Cross-reference authoritative boot-role markers and regression checks; no USER sudo enablement.

Family	Exact technical scope	A/B/C evidence IDs
03.01	USER, SYSMAINT and root account separation	KHAHP-03.01A · KHAHP-03.01B · KHAHP-03.01C
03.02	Distinct UID/GID and home-directory boundaries	KHAHP-03.02A · KHAHP-03.02B · KHAHP-03.02C
03.03	user-sysmaint-split package and owned policy files	KHAHP-03.03A · KHAHP-03.03B · KHAHP-03.03C
03.04	sudo binary ownership, mode and SYSMAINT-only execution	KHAHP-03.04A · KHAHP-03.04B · KHAHP-03.04C
03.05	sudoers syntax and user-sysmaint- split policy	KHAHP-03.05A · KHAHP-03.05B · KHAHP-03.05C
03.06	USER negative sudo/root acquisition probe	KHAHP-03.06A · KHAHP-03.06B · KHAHP-03.06C
03.07	Login-shell and boot-role separation behavior	KHAHP-03.07A · KHAHP-03.07B · KHAHP-03.07C

04 — SYSMAINT Boot Role

Observed implementation/evidence: SYSMAINT evidence was collected in a privileged maintenance session; normal USER boot retained ordinary browser and network functionality.

Acceptance boundary / next proof: Tie protected role markers, role-specific units and provenance to current release acceptance.

Family	Exact technical scope	A/B/C evidence IDs
04.01	boot-role=sysmaint kernel marker	KHAHP-04.01A · KHAHP-04.01B · KHAHP-04.01C
04.02	systemd.unit=sysmaint-boot.target marker	KHAHP-04.02A · KHAHP-04.02B · KHAHP-04.02C
04.03	sysmaint-boot.target package provenance and unit content	KHAHP-04.03A · KHAHP-04.03B · KHAHP-04.03C
04.04	Required SYSMAINT dependencies and wanted units	KHAHP-04.04A · KHAHP-04.04B · KHAHP-04.04C
04.05	Intentionally omitted USER-only or incompatible hardeners	KHAHP-04.05A · KHAHP-04.05B · KHAHP-04.05C
04.06	Conflicting USER/graphical boot-role negative checks	KHAHP-04.06A · KHAHP-04.06B · KHAHP-04.06C
04.07	Failed units and SYSMAINT boot health	KHAHP-04.07A · KHAHP-04.07B · KHAHP-04.07C

05 — Accounts / UID / GID / Local Identity

Observed implementation/evidence: Restricted USER account and service-account inventory were reviewed; no arbitrary account changes were justified.

Acceptance boundary / next proof: Document the `sync` service-account classification and identity persistence.

Historical action references: U01-004.

Family	Exact technical scope	A/B/C evidence IDs
05.01	root, USER and SYSMAINT account identity	KHAHP-05.01A · KHAHP-05.01B · KHAHP-05.01C
05.02	Duplicate UID and duplicate GID detection	KHAHP-05.02A · KHAHP-05.02B · KHAHP-05.02C
05.03	UID 0 exclusivity and privileged- account review	KHAHP-05.03A · KHAHP-05.03B · KHAHP-05.03C
05.04	Low-UID service-account shell classification	KHAHP-05.04A · KHAHP-05.04B · KHAHP-05.04C
05.05	Home-directory ownership and permissions	KHAHP-05.05A · KHAHP-05.05B · KHAHP-05.05C
05.06	shadow/gshadow/passwd/group ownership and modes	KHAHP-05.06A · KHAHP-05.06B · KHAHP-05.06C
05.07	NSS identity source configuration and root lock state	KHAHP-05.07A · KHAHP-05.07B · KHAHP-05.07C

06 — LUKS / Disk / Swap Encryption

Observed implementation/evidence: LUKS-encrypted root block mapping was observed; the removable installation/recovery disk was kept separate from system storage.

Acceptance boundary / next proof: Private key/header backup, encrypted backup and isolated restore demonstration remain separate acceptance requirements.

Historical action references: U01-009.

Family	Exact technical scope	A/B/C evidence IDs
06.01	Root filesystem source and dm-crypt mapping	KHAHP-06.01A · KHAHP-06.01B · KHAHP-06.01C
06.02	LUKS2 backing partition and cryptsetup status	KHAHP-06.02A · KHAHP-06.02B · KHAHP-06.02C
06.03	LUKS header metadata without key- material disclosure	KHAHP-06.03A · KHAHP-06.03B · KHAHP-06.03C
06.04	Root encryption chain from block device to mountpoint	KHAHP-06.04A · KHAHP-06.04B · KHAHP-06.04C
06.05	Active swap inventory and encrypted placement	KHAHP-06.05A · KHAHP-06.05B · KHAHP-06.05C
06.06	swapfile ownership, mode and containing filesystem	KHAHP-06.06A · KHAHP-06.06B · KHAHP-06.06C
06.07	crypttab/fstab encryption persistence and boot mapping	KHAHP-06.07A · KHAHP-06.07B · KHAHP-06.07C

07 — Filesystem / Mount Security

Observed implementation/evidence: A controlled earlier change set `/tmp` to `nodev,nosuid`. On October 18 the `/boot` entry was placed before `/boot/efi`; privileged `findmnt --verify --verbose` reported no warnings or errors, with a timestamped backup. `noexec` remains deliberately excluded.

Acceptance boundary / next proof: Capture next approved post-reboot mount-order persistence and application compatibility; do not apply global noexec merely to match a benchmark.

Historical action references: `U01-001`, `U01-002`, `U01-011`, `U01-012`, `U01-014`.

Family	Exact technical scope	A/B/C evidence IDs
07.01	Effective mount table and filesystem inventory	KHAHP-07.01A · KHAHP-07.01B · KHAHP-07.01C
07.02	Root mount source, filesystem and options	KHAHP-07.02A · KHAHP-07.02B · KHAHP-07.02C
07.03	/boot and /boot/efi separation, ownership and exposure	KHAHP-07.03A · KHAHP-07.03B · KHAHP-07.03C
07.04	/tmp tmpfs, sticky bit and mount- option review	KHAHP-07.04A · KHAHP-07.04B · KHAHP-07.04C
07.05	/var/tmp sticky-bit and separate- mount review	KHAHP-07.05A · KHAHP-07.05B · KHAHP-07.05C
07.06	/home containment, ownership and inherited options	KHAHP-07.06A · KHAHP-07.06B · KHAHP-07.06C
07.07	Writable mountpoints, fstab and remount-secure review	KHAHP-07.07A · KHAHP-07.07B · KHAHP-07.07C

08 — GRUB / Bootloader Protection

Observed implementation/evidence: GRUB protection directives were observed in the historical technical review.

Acceptance boundary / next proof: Physical unauthorized edit-denial and protected recovery proof have not been supplied; no secret disclosure.

Historical action references: `U01-008`.

Family	Exact technical scope	A/B/C evidence IDs
08.01	GRUB package baseline and generated grub.cfg	KHAHP-08.01A · KHAHP-08.01B · KHAHP-08.01C
08.02	grub.cfg ownership, permissions and menu entries	KHAHP-08.02A · KHAHP-08.02B · KHAHP-08.02C
08.03	GRUB password_pbkdf2 and superusers directives	KHAHP-08.03A · KHAHP-08.03B · KHAHP-08.03C

Family	Exact technical scope	A/B/C evidence IDs
08.04	Kicksecure CPU/kernel/ram-wipe/remount/signed -module fragments	KHAHP-08.04A · KHAHP-08.04B · KHAHP-08.04C
08.05	Quiet-boot and recovery-restriction fragments	KHAHP-08.05A · KHAHP-08.05B · KHAHP-08.05C
08.06	Effective boot command-line correlation	KHAHP-08.06A · KHAHP-08.06B · KHAHP-08.06C
08.07	/boot and EFI exposure versus cryptographic integrity limits	KHAHP-08.07A · KHAHP-08.07B · KHAHP-08.07C

09 — CPU Vulnerability Mitigations

Observed implementation/evidence: Spectre-class, SMT and memory-mitigation kernel options were observed; boot parameters alone are not proof of every mitigation.

Acceptance boundary / next proof: Correlate kernel capability, microcode and runtime vulnerabilities rather than counting cmdline tokens.

Family	Exact technical scope	A/B/C evidence IDs
09.01	CPU identity, vendor and mitigation- capability baseline	KHAHP-09.01A · KHAHP-09.01B · KHAHP-09.01C
09.02	SMT/Hyper-Threading disablement and topology	KHAHP-09.02A · KHAHP-09.02B · KHAHP-09.02C
09.03	Runtime microcode revision and package baseline	KHAHP-09.03A · KHAHP-09.03B · KHAHP-09.03C
09.04	Kernel vulnerability-interface enumeration	KHAHP-09.04A · KHAHP-09.04B · KHAHP-09.04C
09.05	Spectre/Meltdown/L1TF/MDS/ Retbleed/BHI/SSB/TAA/GDS mitigations	KHAHP-09.05A · KHAHP-09.05B · KHAHP-09.05C
09.06	Mitigation-related kernel command- line parameters	KHAHP-09.06A · KHAHP-09.06B · KHAHP-09.06C
09.07	CPU mitigation flags and kernel-log confirmation	KHAHP-09.07A · KHAHP-09.07B · KHAHP-09.07C

10 — IOMMU / DMA Protection

Observed implementation/evidence: The SYSMAINT evidence reported 17 IOMMU groups and enabled Secure Boot.

Acceptance boundary / next proof: Firmware/preboot DMA applicability remains a separate hardware/recovery question; owner BIOS work is not reopened.

Historical action references: U01-010.

Family	Exact technical scope	A/B/C evidence IDs
10.01	CPU virtualization/VT-d capability baseline	KHAHP-10.01A · KHAHP-10.01B · KHAHP-10.01C
10.02	IOMMU kernel parameters and passthrough policy	KHAHP-10.02A · KHAHP-10.02B · KHAHP-10.02C
10.03	DMAR/IOMMU runtime enablement	KHAHP-10.03A · KHAHP-10.03B · KHAHP-10.03C
10.04	Translated default domain and strict invalidation	KHAHP-10.04A · KHAHP-10.04B · KHAHP-10.04C
10.05	IRQ/interrupt remapping	KHAHP-10.05A · KHAHP-10.05B · KHAHP-10.05C
10.06	IOMMU group topology and full PCI membership	KHAHP-10.06A · KHAHP-10.06B · KHAHP-10.06C
10.07	Early PCI DMA restriction and firmware WSMT evidence	KHAHP-10.07A · KHAHP-10.07B · KHAHP-10.07C

11 — Kernel Sysctl Hardening

Observed implementation/evidence: `kernel.kptr_restrict=2`, `dmesg_restrict=1`, `ptrace_scope=3`, unprivileged BPF restrictions and protected file operations were observed.

Acceptance boundary / next proof: Verify package precedence and post-upgrade persistence where material.

Family	Exact technical scope	A/B/C evidence IDs
11.01	Sysctl source-tree inventory and precedence	KHAHP-11.01A · KHAHP-11.01B · KHAHP-11.01C
11.02	Kicksecure security-misc sysctl source files	KHAHP-11.02A · KHAHP-11.02B · KHAHP-11.02C
11.03	Kernel information-leak restrictions	KHAHP-11.03A · KHAHP-11.03B · KHAHP-11.03C
11.04	ASLR, mmap, kexec and privileged- coredump controls	KHAHP-11.04A · KHAHP-11.04B · KHAHP-11.04C
11.05	BPF, ptrace, perf and TTY line- discipline restrictions	KHAHP-11.05A · KHAHP-11.05B · KHAHP-11.05C
11.06	Protected links/FIFOs/regular-file controls	KHAHP-11.06A · KHAHP-11.06B · KHAHP-11.06C
11.07	Network-stack sysctl hardening observations	KHAHP-11.07A · KHAHP-11.07B · KHAHP-11.07C

12 — Kernel Boot / Memory Hardening

Observed implementation/evidence: Memory and allocator hardening boot flags were observed in the effective command line.

Acceptance boundary / next proof: Do not promote CFI/KCFI command-line tokens into confirmed kernel enforcement without build/runtime proof.

Historical action references: U01-005.

Family	Exact technical scope	A/B/C evidence IDs
12.01	Slab allocator hardening and slab_nomerge	KHAHP-12.01A · KHAHP-12.01B · KHAHP-12.01C
12.02	init_on_alloc and init_on_free memory initialization	KHAHP-12.02A · KHAHP-12.02B · KHAHP-12.02C
12.03	page allocator shuffle and stack randomization	KHAHP-12.03A · KHAHP-12.03B · KHAHP-12.03C
12.04	vsyscall, vdso32 and 32-bit compatibility restrictions	KHAHP-12.04A · KHAHP-12.04B · KHAHP-12.04C
12.05	debugfs, KFENCE and kernel debug- surface controls	KHAHP-12.05A · KHAHP-12.05B · KHAHP-12.05C
12.06	CFI/KCFI and boot-time control acceptance	KHAHP-12.06A · KHAHP-12.06B · KHAHP-12.06C
12.07	pstore/ERST/proc_mem/entropy trust boot parameters	KHAHP-12.07A · KHAHP-12.07B · KHAHP-12.07C

13 — Kernel Module Restrictions

Observed implementation/evidence: Kernel/module policy inventories exist from earlier paired-role observations.

Acceptance boundary / next proof: Resolve built-in versus loadable module applicability without loading prohibited test modules.

Historical action references: U01-006.

Family	Exact technical scope	A/B/C evidence IDs
13.01	security-misc module blacklist configuration	KHAHP-13.01A · KHAHP-13.01B · KHAHP-13.01C
13.02	Hard-disable install rules for risky modules	KHAHP-13.02A · KHAHP-13.02B · KHAHP-13.02C
13.03	Loaded-module inventory versus policy	KHAHP-13.03A · KHAHP-13.03B · KHAHP-13.03C
13.04	Legacy filesystem module restrictions	KHAHP-13.04A · KHAHP-13.04B · KHAHP-13.04C
13.05	Legacy network/protocol module restrictions	KHAHP-13.05A · KHAHP-13.05B · KHAHP-13.05C
13.06	USB/storage/media module restriction exceptions	KHAHP-13.06A · KHAHP-13.06B · KHAHP-13.06C
13.07	Module-load negative tests and compatibility exceptions	KHAHP-13.07A · KHAHP-13.07B · KHAHP-13.07C

14 — Kernel Module Signing / DKMS

Observed implementation/evidence: Privileged evidence identified signed Tirdad DKMS builds; Secure Boot was enabled in the October 18 SYSMAINT check.

Acceptance boundary / next proof: Signed-module rejection, future kernel-update regression and key-provenance acceptance require controlled proof.

Historical action references: `U01-007`, `U01-013`.

Family	Exact technical scope	A/B/C evidence IDs
`14.01`	Loaded-module signer inventory	`KHAHP-14.01A` · `KHAHP-14.01B` · `KHAHP-14.01C`
`14.02`	DKMS package and built-module inventory	`KHAHP-14.02A` · `KHAHP-14.02B` · `KHAHP-14.02C`
`14.03`	tirdad DKMS build, signer and runtime state	`KHAHP-14.03A` · `KHAHP-14.03B` · `KHAHP-14.03C`
`14.04`	NVIDIA/other third-party module signer state	`KHAHP-14.04A` · `KHAHP-14.04B` · `KHAHP-14.04C`
`14.05`	Kernel module signature enforcement capability	`KHAHP-14.05A` · `KHAHP-14.05B` · `KHAHP-14.05C`
`14.06`	MOK/enrolled certificate readiness	`KHAHP-14.06A` · `KHAHP-14.06B` · `KHAHP-14.06C`
`14.07`	Secure-Boot compatibility and unsigned-module exceptions	`KHAHP-14.07A` · `KHAHP-14.07B` · `KHAHP-14.07C`

15 — AppArmor / Mandatory Access Control

Observed implementation/evidence: AppArmor is enabled with many loaded profiles; Firefox 158 uses a vendor-owned `flags=(unconfined)` placeholder, and USER browser processes were observed `firefox (unconfined)`. Selected Firefox processes did have seccomp filters.

Acceptance boundary / next proof: Full restrictive Firefox confinement is not implemented: prepare an isolated compatible profile and test complain/enforce transitions with rollback; do not `aa-enforce` the placeholder.

Historical action references: `U15-001`, `U15-002`, `U15-003`.

Family	Exact technical scope	A/B/C evidence IDs
`15.01`	AppArmor kernel/module enablement	`KHAHP-15.01A` · `KHAHP-15.01B` · `KHAHP-15.01C`
`15.02`	Profile inventory and parser status	`KHAHP-15.02A` · `KHAHP-15.02B` · `KHAHP-15.02C`
`15.03`	Enforce/complain/unconfined process counts	`KHAHP-15.03A` · `KHAHP-15.03B` · `KHAHP-15.03C`
`15.04`	Tor and network-service profile enforcement	`KHAHP-15.04A` · `KHAHP-15.04B` · `KHAHP-15.04C`

Family	Exact technical scope	A/B/C evidence IDs
15.05	Browser/application profile coverage	KHAHP-15.05A · KHAHP-15.05B · KHAHP-15.05C
15.06	Denied-event review and false-positive handling	KHAHP-15.06A · KHAHP-15.06B · KHAHP-15.06C
15.07	USER versus SYSMAINT AppArmor behavior	KHAHP-15.07A · KHAHP-15.07B · KHAHP-15.07C

16 — SUID / SGID / File Capabilities

Observed implementation/evidence: SUID/SGID and capabilities were inventoried. No evidence justifies blanket removal of package-owned helper capabilities.

Acceptance boundary / next proof: Resolve privileged-binary drift and capability-specific exception decisions.

Historical action references: U15-004, U15-005.

Family	Exact technical scope	A/B/C evidence IDs
16.01	Complete SUID file inventory	KHAHP-16.01A · KHAHP-16.01B · KHAHP-16.01C
16.02	Complete SGID file inventory	KHAHP-16.02A · KHAHP-16.02B · KHAHP-16.02C
16.03	Package ownership of privileged binaries	KHAHP-16.03A · KHAHP-16.03B · KHAHP-16.03C
16.04	permission-hardener expected mode validation	KHAHP-16.04A · KHAHP-16.04B · KHAHP-16.04C
16.05	File-capability inventory and package attribution	KHAHP-16.05A · KHAHP-16.05B · KHAHP-16.05C
16.06	Unexpected privileged executable detection	KHAHP-16.06A · KHAHP-16.06B · KHAHP-16.06C
16.07	Privilege-binary negative/compatibility review	KHAHP-16.07A · KHAHP-16.07B · KHAHP-16.07C

17 — Permissions / ACL / Umask

Observed implementation/evidence: Restrictive file- and temporary-directory controls were observed. A vendor writable-status exception is documented.

Acceptance boundary / next proof: Accept or reject vendor-permission exceptions on evidence; test post-update drift.

Historical action references: U15-006, U15-007.

Family	Exact technical scope	A/B/C evidence IDs
17.01	System file ownership baseline	KHAHP-17.01A · KHAHP-17.01B · KHAHP-17.01C
17.02	World-writable system path inventory	KHAHP-17.02A · KHAHP-17.02B · KHAHP-17.02C
17.03	Unexpected group-writable sensitive paths	KHAHP-17.03A · KHAHP-17.03B · KHAHP-17.03C
17.04	ACL inventory on sensitive configuration	KHAHP-17.04A · KHAHP-17.04B · KHAHP-17.04C
17.05	Root and user umask policy	KHAHP-17.05A · KHAHP-17.05B · KHAHP-17.05C
17.06	Home and evidence-directory permissions	KHAHP-17.06A · KHAHP-17.06B · KHAHP-17.06C
17.07	Sticky-bit and shared-directory permission policy	KHAHP-17.07A · KHAHP-17.07B · KHAHP-17.07C

18 — PAM / Login / Faillock

Observed implementation/evidence: Yescrypt/PAM hardening was observed. The SYSMAINT review found `faillock deny=50` and `unlock_time=never`.

Acceptance boundary / next proof: Document safe lockout recovery and approved password/aging choices without deliberate account lockout.

Historical action references: `U15-008`, `U15-009`, `U15-010`.

Family	Exact technical scope	A/B/C evidence IDs
18.01	PAM stack package provenance	KHAHP-18.01A · KHAHP-18.01B · KHAHP-18.01C
18.02	password quality and hashing policy	KHAHP-18.02A · KHAHP-18.02B · KHAHP-18.02C
18.03	faillock/brute-force lockout policy	KHAHP-18.03A · KHAHP-18.03B · KHAHP-18.03C
18.04	login delay and retry controls	KHAHP-18.04A · KHAHP-18.04B · KHAHP-18.04C
18.05	su/sudo/polkit PAM interaction	KHAHP-18.05A · KHAHP-18.05B · KHAHP-18.05C
18.06	account aging and inactive-account policy	KHAHP-18.06A · KHAHP-18.06B · KHAHP-18.06C
18.07	USER/SYSMAINT login regression tests	KHAHP-18.07A · KHAHP-18.07B · KHAHP-18.07C

19 — SSH Client / Server Security

Observed implementation/evidence: The latest USER snapshot showed the SSH server inactive and no TCP SSH listener; local client and socket uses are classified separately.

Acceptance boundary / next proof: Confirm listener/role persistence after update without enabling a network SSH daemon.

Historical action references: `U15-011`, `U15-012`.

Family	Exact technical scope	A/B/C evidence IDs
19.01	Local SSH server service/socket state	KHAHP-19.01A · KHAHP-19.01B · KHAHP-19.01C
19.02	sshd listener and inbound port absence	KHAHP-19.02A · KHAHP-19.02B · KHAHP-19.02C
19.03	sshd_config security-misc policy if installed	KHAHP-19.03A · KHAHP-19.03B · KHAHP-19.03C
19.04	SSH client configuration and host-key verification	KHAHP-19.04A · KHAHP-19.04B · KHAHP-19.04C
19.05	Private-key ownership and permission baseline	KHAHP-19.05A · KHAHP-19.05B · KHAHP-19.05C
19.06	Password-authentication and root-login policy for future server use	KHAHP-19.06A · KHAHP-19.06B · KHAHP-19.06C
19.07	Known-hosts/key algorithm and crypto-policy review	KHAHP-19.07A · KHAHP-19.07B · KHAHP-19.07C

20 — Enabled / Running Service Audit

Observed implementation/evidence: Two October 18 health checks found zero failed systemd units.

Acceptance boundary / next proof: Attribute expected services per USER/SYSMAINT role; monitor post-upgrade changes rather than disable vendor units indiscriminately.

Historical action references: `U15-013`, `U15-014`.

Family	Exact technical scope	A/B/C evidence IDs
20.01	Enabled systemd service inventory	KHAHP-20.01A · KHAHP-20.01B · KHAHP-20.01C
20.02	Running systemd service inventory	KHAHP-20.02A · KHAHP-20.02B · KHAHP-20.02C
20.03	Socket/path/timer activated service inventory	KHAHP-20.03A · KHAHP-20.03B · KHAHP-20.03C
20.04	Kicksecure-required service attribution	KHAHP-20.04A · KHAHP-20.04B · KHAHP-20.04C
20.05	Unexpected or unnecessary service classification	KHAHP-20.05A · KHAHP-20.05B · KHAHP-20.05C
20.06	Listener-to-service ownership correlation	KHAHP-20.06A · KHAHP-20.06B · KHAHP-20.06C
20.07	Reboot persistence and failed-service review	KHAHP-20.07A · KHAHP-20.07B · KHAHP-20.07C

21 — systemd Service Sandboxing

Observed implementation/evidence: Existing systemd sandboxing is present; exposure scores are not themselves exploitable vulnerabilities.

Acceptance boundary / next proof: Evaluate any optional resolver restrictions in a test build with DNS and boot-role regression.

Historical action references: `U15-015`, `U15-016`.

Family	Exact technical scope	A/B/C evidence IDs
21.01	systemd-analyze security baseline	KHAHP-21.01A · KHAHP-21.01B · KHAHP-21.01C
21.02	NoNewPrivileges adoption	KHAHP-21.02A · KHAHP-21.02B · KHAHP-21.02C
21.03	ProtectSystem/ProtectHome restrictions	KHAHP-21.03A · KHAHP-21.03B · KHAHP-21.03C
21.04	PrivateTmp/PrivateDevices isolation	KHAHP-21.04A · KHAHP-21.04B · KHAHP-21.04C
21.05	CapabilityBoundingSet and ambient capabilities	KHAHP-21.05A · KHAHP-21.05B · KHAHP-21.05C
21.06	RestrictNamespaces/SystemCallFilter/address-family controls	KHAHP-21.06A · KHAHP-21.06B · KHAHP-21.06C
21.07	Service-specific compatibility exceptions and regressions	KHAHP-21.07A · KHAHP-21.07B · KHAHP-21.07C

22 — System.map / Core Dumps / PStore

Observed implementation/evidence: Crash-dump/pstore suppression controls and associated privacy-versus-forensics tradeoffs were documented.

Acceptance boundary / next proof: Record exception rationale and observe package/kernel update drift; do not generate a production kernel panic.

Historical action references: `U15-017`, `U15-018`.

Family	Exact technical scope	A/B/C evidence IDs
22.01	System.map exposure and removal policy	KHAHP-22.01A · KHAHP-22.01B · KHAHP-22.01C
22.02	systemd-coredump configuration	KHAHP-22.02A · KHAHP-22.02B · KHAHP-22.02C
22.03	kernel.core_pattern effective state	KHAHP-22.03A · KHAHP-22.03B · KHAHP-22.03C
22.04	fs.suid_dumpable effective state	KHAHP-22.04A · KHAHP-22.04B · KHAHP-22.04C
22.05	pstore retention/disable policy	KHAHP-22.05A · KHAHP-22.05B · KHAHP-22.05C
22.06	EFI pstore and ERST boot controls	KHAHP-22.06A · KHAHP-22.06B · KHAHP-22.06C
22.07	Crash-evidence privacy versus diagnostic requirements	KHAHP-22.07A · KHAHP-22.07B · KHAHP-22.07C

23 — Journald / Logging / Logrotate

Observed implementation/evidence: Journald/secure-time baseline checks exist. Logging integrity is not equivalent to complete audit rule coverage.

Acceptance boundary / next proof: Approve retention and forwarding rules, ensure time coherence and protect evidence custody.

Historical action references: `U15-019`, `U15-020`.

Family	Exact technical scope	A/B/C evidence IDs
23.01	journald security-misc configuration	KHAHP-23.01A · KHAHP-23.01B · KHAHP-23.01C
23.02	Persistent versus volatile journal storage	KHAHP-23.02A · KHAHP-23.02B · KHAHP-23.02C
23.03	Journal file permissions and ownership	KHAHP-23.03A · KHAHP-23.03B · KHAHP-23.03C
23.04	Rate limiting and size/retention policy	KHAHP-23.04A · KHAHP-23.04B · KHAHP-23.04C

Family	Exact technical scope	A/B/C evidence IDs
23.05	Forwarding/syslog integration state	KHAHP-23.05A · KHAHP-23.05B · KHAHP-23.05C
23.06	logrotate configuration and sensitive-log modes	KHAHP-23.06A · KHAHP-23.06B · KHAHP-23.06C
23.07	Log integrity, time coherence and reboot persistence	KHAHP-23.07A · KHAHP-23.07B · KHAHP-23.07C

24 — Panic / Recovery / Emergency Shutdown

Observed implementation/evidence: Emergency/recovery restrictions and shutdown service behavior were inventoried.

Acceptance boundary / next proof: Use disposable media/VM for fault injection; do not trigger an unexpected live-system panic.

Historical action references: U15-021, U15-022.

Family	Exact technical scope	A/B/C evidence IDs
24.01	kernel panic-on-oops policy	KHAHP-24.01A · KHAHP-24.01B · KHAHP-24.01C
24.02	panic timeout and fail-closed behavior	KHAHP-24.02A · KHAHP-24.02B · KHAHP-24.02C
24.03	emerg-shutdown.service provenance and state	KHAHP-24.03A · KHAHP-24.03B · KHAHP-24.03C
24.04	rescue/emergency target restrictions	KHAHP-24.04A · KHAHP-24.04B · KHAHP-24.04C
24.05	rd.emergency=halt and rd.shell=0 behavior	KHAHP-24.05A · KHAHP-24.05B · KHAHP-24.05C
24.06	Magic SysRq and recovery-surface policy	KHAHP-24.06A · KHAHP-24.06B · KHAHP-24.06C
24.07	Controlled failure/recovery evidence and rollback	KHAHP-24.07A · KHAHP-24.07B · KHAHP-24.07C

25 — RAM Wipe

Observed implementation/evidence: Kernel memory initialization controls and encrypted storage mapping are observed.

Acceptance boundary / next proof: Verify wipe-hook lifecycle and swap/crypt ancestry without claiming immunity to cold-boot attacks.

Historical action references: U25-003, U25-004, U25-017.

Family	Exact technical scope	A/B/C evidence IDs
25.01	ram-wipe package provenance and version	KHAHP-25.01A · KHAHP-25.01B · KHAHP-25.01C
25.02	GRUB/package RAM-wipe source presence	KHAHP-25.02A · KHAHP-25.02B · KHAHP-25.02C
25.03	init_on_alloc/init_on_free live correlation	KHAHP-25.03A · KHAHP-25.03B · KHAHP-25.03C
25.04	Shutdown/reboot wipe hook inventory	KHAHP-25.04A · KHAHP-25.04B · KHAHP-25.04C
25.05	Swap/LUKS interaction during shutdown	KHAHP-25.05A · KHAHP-25.05B · KHAHP-25.05C

Family	Exact technical scope	A/B/C evidence IDs
25.06	Controlled reboot/shutdown persistence test	KHAHP-25.06A · KHAHP-25.06B · KHAHP-25.06C
25.07	Limitations of software RAM wiping and cold-boot scope	KHAHP-25.07A · KHAHP-25.07B · KHAHP-25.07C

26 — USBGuard / Peripheral Security

Observed implementation/evidence: USBGuard was active and enabled; its policy authorized a recognized ZOWIE mouse and SanDisk removable medium. Approved USB mice, keyboards and recovery storage are allowed; a webcam was USB-authorized but its USER video device nodes denied direct access.

Acceptance boundary / next proof: Second mouse, external-keyboard functional acceptance, phone/network composite negative testing and policy precedence still need controlled proof.

Historical action references: U25-001, U25-002, U25-018.

Family	Exact technical scope	A/B/C evidence IDs
26.01	usbguard daemon and DBus service state	KHAHP-26.01A · KHAHP-26.01B · KHAHP-26.01C
26.02	Policy files and rule provenance	KHAHP-26.02A · KHAHP-26.02B · KHAHP-26.02C
26.03	ImplicitPolicyTarget and device-policy settings	KHAHP-26.03A · KHAHP-26.03B · KHAHP-26.03C
26.04	Allowed keyboard/mouse/audio/video/storage classes	KHAHP-26.04A · KHAHP-26.04B · KHAHP-26.04C
26.05	Rejected HID+extra and RNDIS/network-device patterns	KHAHP-26.05A · KHAHP-26.05B · KHAHP-26.05C
26.06	IPC authorization/notifier behavior	KHAHP-26.06A · KHAHP-26.06B · KHAHP-26.06C
26.07	Hot-plug negative tests and reboot persistence	KHAHP-26.07A · KHAHP-26.07B · KHAHP-26.07C

27 — NetworkManager / Wi-Fi / MAC / DHCP

Observed implementation/evidence: NetworkManager uses a Wi-Fi connection and randomized identity controls were historically configured.

Acceptance boundary / next proof: Over-the-air MAC and DHCP identity comparison on an owned router remains a dedicated test.

Historical action references: U25-005, U25-006.

Family	Exact technical scope	A/B/C evidence IDs
27.01	NetworkManager package/service state	KHAHP-27.01A · KHAHP-27.01B · KHAHP-27.01C
27.02	Saved Wi-Fi profile inventory and autoconnect	KHAHP-27.02A · KHAHP-27.02B · KHAHP-27.02C
27.03	Active-profile cloned-MAC and DHCP privacy settings	KHAHP-27.03A · KHAHP-27.03B · KHAHP-27.03C
27.04	Scan MAC randomization	KHAHP-27.04A · KHAHP-27.04B · KHAHP-27.04C
27.05	Ethernet MAC randomization policy	KHAHP-27.05A · KHAHP-27.05B · KHAHP-27.05C
27.06	Runtime MAC versus permanent hardware MAC	KHAHP-27.06A · KHAHP-27.06B · KHAHP-27.06C
27.07	DHCP client identifier and profile persistence	KHAHP-27.07A · KHAHP-27.07B · KHAHP-27.07C

28 — IPv6 Policy

Observed implementation/evidence: The inspected kernel boot used `ipv6.disable=1`, and the USER network snapshot showed no IPv6 route.

Acceptance boundary / next proof: Explicitly decide whether to retain disabled IPv6 or move to a separately reviewed dual-stack privacy architecture.

Historical action references: `U25-007`, `U25-019`.

Family	Exact technical scope	A/B/C evidence IDs
28.01	ipv6.disable=1 boot parameter	KHAHP-28.01A · KHAHP-28.01B · KHAHP-28.01C
28.02	Absence of IPv6 addresses	KHAHP-28.02A · KHAHP-28.02B · KHAHP-28.02C
28.03	Absence of IPv6 routes	KHAHP-28.03A · KHAHP-28.03B · KHAHP-28.03C
28.04	NetworkManager IPv6 method/state	KHAHP-28.04A · KHAHP-28.04B · KHAHP-28.04C
28.05	Resolver absence of IPv6 nameservers	KHAHP-28.05A · KHAHP-28.05B · KHAHP-28.05C
28.06	IPv6 TCP/UDP listener exposure	KHAHP-28.06A · KHAHP-28.06B · KHAHP-28.06C
28.07	Reboot persistence and DNS/HTTPS regression test	KHAHP-28.07A · KHAHP-28.07B · KHAHP-28.07C

29 — Unbound / Local DNS Architecture

Observed implementation/evidence: Unbound is active and configured, and the USER snapshot showed a `127.0.0.1:53` listener; local DNS resolution was functional.

Acceptance boundary / next proof: Distinguish effective system resolver wiring from application bypass during network transitions.

Historical action references: `U25-008`, `U25-020`.

Family	Exact technical scope	A/B/C evidence IDs
29.01	Unbound package/service state	KHAHP-29.01A · KHAHP-29.01B · KHAHP-29.01C
29.02	127.0.0.1:53 UDP/TCP listener and non-loopback absence	KHAHP-29.02A · KHAHP-29.02B · KHAHP-29.02C
29.03	resolv.conf local-resolver wiring	KHAHP-29.03A · KHAHP-29.03B · KHAHP-29.03C
29.04	NetworkManager effective DNS assignment	KHAHP-29.04A · KHAHP-29.04B · KHAHP-29.04C
29.05	Unbound configuration include tree validation	KHAHP-29.05A · KHAHP-29.05B · KHAHP-29.05C
29.06	DoT forward-zone and IPv6 source policy	KHAHP-29.06A · KHAHP-29.06B · KHAHP-29.06C
29.07	USER/SYSMAINT service behavior and persistence	KHAHP-29.07A · KHAHP-29.07B · KHAHP-29.07C

30 — DNSSEC / DoT / DNS Leak Prevention

Observed implementation/evidence: A DNSSEC-valid response set the AD flag; a deliberately invalid DNSSEC domain returned SERVFAIL. Unbound DoT forwarding and a port-853 process observation were previously reported.

Acceptance boundary / next proof: Certificate-negative transport validation and bounded external UDP/TCP 53 bypass verdicts are not yet signed off.

Historical action references: U25-009, U25-010, U25-011, U25-021.

Family	Exact technical scope	A/B/C evidence IDs
30.01	DNSSEC positive validation test	KHAHP-30.01A · KHAHP-30.01B · KHAHP-30.01C
30.02	Broken-DNSSEC negative test	KHAHP-30.02A · KHAHP-30.02B · KHAHP-30.02C
30.03	DoT forwarding configuration and reachability scope	KHAHP-30.03A · KHAHP-30.03B · KHAHP-30.03C
30.04	Plaintext external UDP/53 block	KHAHP-30.04A · KHAHP-30.04B · KHAHP-30.04C
30.05	Plaintext external TCP/53 block	KHAHP-30.05A · KHAHP-30.05B · KHAHP-30.05C
30.06	Local DNS functionality after firewall enforcement	KHAHP-30.06A · KHAHP-30.06B · KHAHP-30.06C
30.07	DNS leak/bypass and reboot-persistence testing	KHAHP-30.07A · KHAHP-30.07B · KHAHP-30.07C

31 — nftables Firewall

Observed implementation/evidence: nftables is active; privileged rules show input and forward default-drop policies, but the OUTPUT base policy permits traffic with explicit direct-DNS restrictions.

Acceptance boundary / next proof: Correlate final effective OUTPUT rule order and UDP/TCP DNS verdicts at an authorized endpoint; verify normalized reboot persistence.

Historical action references: `U25-012`, `U25-013`.

Family	Exact technical scope	A/B/C evidence IDs
31.01	nftables package/service/runtime ruleset state	KHAHP-31.01A · KHAHP-31.01B · KHAHP-31.01C
31.02	Dedicated inet kicksecure_hardening table	KHAHP-31.02A · KHAHP-31.02B · KHAHP-31.02C
31.03	Input default-drop policy	KHAHP-31.03A · KHAHP-31.03B · KHAHP-31.03C
31.04	Forward default-drop policy	KHAHP-31.04A · KHAHP-31.04B · KHAHP-31.04C
31.05	Loopback, established/related and invalid handling	KHAHP-31.05A · KHAHP-31.05B · KHAHP-31.05C
31.06	DHCP/ICMP and required inbound exceptions	KHAHP-31.06A · KHAHP-31.06B · KHAHP-31.06C
31.07	Static syntax, reload safety and reboot persistence	KHAHP-31.07A · KHAHP-31.07B · KHAHP-31.07C

32 — LAN Discovery / Multicast / LLMNR / mDNS

Observed implementation/evidence: LAN discovery service and listener inventory was collected.

Acceptance boundary / next proof: Peer-origin mDNS, LLMNR and SSDP reply-negative tests remain inconclusive without an owned LAN peer.

Historical action references: `U25-014`, `U25-022`.

Family	Exact technical scope	A/B/C evidence IDs
32.01	Avahi/mDNS daemon state	KHAHP-32.01A · KHAHP-32.01B · KHAHP-32.01C
32.02	LLMNR/resolved multicast name-resolution state	KHAHP-32.02A · KHAHP-32.02B · KHAHP-32.02C
32.03	wsdd/Samba discovery state	KHAHP-32.03A · KHAHP-32.03B · KHAHP-32.03C
32.04	rpcbind/minidlna/UPnP service state	KHAHP-32.04A · KHAHP-32.04B · KHAHP-32.04C
32.05	Multicast/discovery TCP+UDP listener inventory	KHAHP-32.05A · KHAHP-32.05B · KHAHP-32.05C
32.06	Firewall exposure of discovery ports	KHAHP-32.06A · KHAHP-32.06B · KHAHP-32.06C
32.07	Negative LAN discovery and reboot-persistence test	KHAHP-32.07A · KHAHP-32.07B · KHAHP-32.07C

33 — Tor / SOCKS / DNS Isolation

Observed implementation/evidence: Tor was active, with a loopback SOCKS listener observed on port 9050.

Acceptance boundary / next proof: Prove routing and fail-closed behavior for only the applications intended to use Tor; do not stop system-wide services casually.

Historical action references: `U25-015`.

Family	Exact technical scope	A/B/C evidence IDs
33.01	Tor package/service state	KHAHP-33.01A · KHAHP-33.01B · KHAHP-33.01C
33.02	SOCKS listener 127.0.0.1:9050 and non-loopback absence	KHAHP-33.02A · KHAHP-33.02B · KHAHP-33.02C
33.03	Tor configuration provenance and syntax validation	KHAHP-33.03A · KHAHP-33.03B · KHAHP-33.03C
33.04	Application proxy routing where intended	KHAHP-33.04A · KHAHP-33.04B · KHAHP-33.04C
33.05	Tor DNS isolation behavior	KHAHP-33.05A · KHAHP-33.05B · KHAHP-33.05C
33.06	Clearnet bypass classification for Tor-mode applications	KHAHP-33.06A · KHAHP-33.06B · KHAHP-33.06C
33.07	Bootstrap, circuit, restart and failure-mode behavior	KHAHP-33.07A · KHAHP-33.07B · KHAHP-33.07C

34 — sdwdate / Secure Time / Boot Clock

Observed implementation/evidence: sdwdate, boot-time and secure-time components were observed in the historical baseline.

Acceptance boundary / next proof: Use time/log chronology and disposable-VM rollback tests rather than altering the live system clock.

Historical action references: `U25-016`, `U25-023`.

Family	Exact technical scope	A/B/C evidence IDs
34.01	sdwdate package/service state	KHAHP-34.01A · KHAHP-34.01B · KHAHP-34.01C
34.02	timesanitycheck execution state	KHAHP-34.02A · KHAHP-34.02B · KHAHP-34.02C
34.03	bootclockrandomization execution state	KHAHP-34.03A · KHAHP-34.03B · KHAHP-34.03C
34.04	System clock sanity metadata	KHAHP-34.04A · KHAHP-34.04B · KHAHP-34.04C
34.05	NTP/systemd-timesyncd conflicts	KHAHP-34.05A · KHAHP-34.05B · KHAHP-34.05C
34.06	USER versus SYSMAINT time-source behavior	KHAHP-34.06A · KHAHP-34.06B · KHAHP-34.06C
34.07	Time rollback/jump detection and log coherence	KHAHP-34.07A · KHAHP-34.07B · KHAHP-34.07C

35 — Entropy / RNG

Observed implementation/evidence: The later USER evidence showed entropy availability and strong kernel randomization settings; RNG services were enabled in the system baseline.

Acceptance boundary / next proof: Cloned-image seed uniqueness is a future release test and not a claim about a released product.

Historical action references: `U35-001`.

Family	Exact technical scope	A/B/C evidence IDs
35.01	Kernel RNG readiness at boot	KHAHP-35.01A · KHAHP-35.01B · KHAHP-35.01C
35.02	haveged package/service state	KHAHP-35.02A · KHAHP-35.02B · KHAHP-35.02C
35.03	jitterentropy-rngd package/service state	KHAHP-35.03A · KHAHP-35.03B · KHAHP-35.03C
35.04	random.trust_cpu boot policy	KHAHP-35.04A · KHAHP-35.04B · KHAHP-35.04C
35.05	random.trust_bootloader boot policy	KHAHP-35.05A · KHAHP-35.05B · KHAHP-35.05C
35.06	Entropy source and /dev/random behavior	KHAHP-35.06A · KHAHP-35.06B · KHAHP-35.06C
35.07	Early-boot entropy persistence and regression evidence	KHAHP-35.07A · KHAHP-35.07B · KHAHP-35.07C

36 — Browser DoH / WebRTC / Telemetry

Observed implementation/evidence: The active Firefox `default-release` configuration was observed with DoH/TRR mode 5, WebRTC disabled and relevant prefetch settings disabled. Enabled extension inventory was empty.

Acceptance boundary / next proof: An effective browser-to-network DNS/DoH bypass test using an authorized endpoint remains distinct from preference-file inspection.

Historical action references: `U35-002`.

Family	Exact technical scope	A/B/C evidence IDs
36.01	Browser DNS-over-HTTPS/TRR setting	KHAHP-36.01A · KHAHP-36.01B · KHAHP-36.01C
36.02	Browser proxy DNS routing behavior	KHAHP-36.02A · KHAHP-36.02B · KHAHP-36.02C
36.03	WebRTC/STUN local/public IP exposure	KHAHP-36.03A · KHAHP-36.03B · KHAHP-36.03C

Family	Exact technical scope	A/B/C evidence IDs
36.04	Telemetry and crash-report settings	KHAHP-36.04A · KHAHP-36.04B · KHAHP-36.04C
36.05	Safe browsing/prediction/background network features	KHAHP-36.05A · KHAHP-36.05B · KHAHP-36.05C
36.06	Extension/add-on inventory and trust	KHAHP-36.06A · KHAHP-36.06B · KHAHP-36.06C
36.07	Browser update, profile persistence and leak tests	KHAHP-36.07A · KHAHP-36.07B · KHAHP-36.07C

37 — Application / File / Metadata Privacy

Observed implementation/evidence:
`org.gnome.desktop.thumbnailers disable-all` changed from false to true, and later USER evidence still read true. PCManFM-Qt thumbnail disabling, a synthetic image test and reboot persistence were observed.

Acceptance boundary / next proof: Third-party application history/thumbnail backends and privacy tradeoffs require explicit acceptance. The earlier apply/verify archive self-hash defect is packaging-only.

Historical action references: `U35-003`.

Family	Exact technical scope	A/B/C evidence IDs
37.01	Desktop search/indexing services	KHAHP-37.01A · KHAHP-37.01B · KHAHP-37.01C
37.02	Thumbnail and preview cache behavior	KHAHP-37.02A · KHAHP-37.02B · KHAHP-37.02C
37.03	Recent-files/history and clipboard retention	KHAHP-37.03A · KHAHP-37.03B · KHAHP-37.03C
37.04	Document/image metadata handling	KHAHP-37.04A · KHAHP-37.04B · KHAHP-37.04C
37.05	Temporary-file and application-cache exposure	KHAHP-37.05A · KHAHP-37.05B · KHAHP-37.05C
37.06	Sandboxed versus unsandboxed application inventory	KHAHP-37.06A · KHAHP-37.06B · KHAHP-37.06C
37.07	Metadata-removal workflow and regression testing	KHAHP-37.07A · KHAHP-37.07B · KHAHP-37.07C

38 — Camera / Microphone / Power / Screen Lock

Observed implementation/evidence: Normal USER opening `/dev/video0`, `/dev/video1` and `/dev/snd/pcmC0D0c` was denied without capture. Bluetooth was rfkill-soft-blocked with inactive service. `swaylock` successfully locked the desktop in a shorter manual test.

Acceptance boundary / next proof: Five-minute unattended lock persistence, lid/resume authentication (if policy requires it), keyboard/mouse hotplug and hibernation decision remain distinct tests.

Historical action references: `U35-004`.

Family	Exact technical scope	A/B/C evidence IDs
38.01	Camera hardware/device exposure	KHAHP-38.01A · KHAHP-38.01B · KHAHP-38.01C
38.02	Microphone hardware/device exposure	KHAHP-38.02A · KHAHP-38.02B · KHAHP-38.02C
38.03	Bluetooth/rfkill interaction with peripherals	KHAHP-38.03A · KHAHP-38.03B · KHAHP-38.03C
38.04	Screen-lock timeout and authentication	KHAHP-38.04A · KHAHP-38.04B · KHAHP-38.04C
38.05	Suspend/lock-on-lid and idle behavior	KHAHP-38.05A · KHAHP-38.05B · KHAHP-38.05C
38.06	Hibernation configuration and encrypted-swap implications	KHAHP-38.06A · KHAHP-38.06B · KHAHP-38.06C
38.07	Shutdown-versus-suspend operational policy	KHAHP-38.07A · KHAHP-38.07B · KHAHP-38.07C

39 — Firmware / BIOS / Secure Boot / TPM

Observed implementation/evidence: OWNER COMPLETED / EXCLUDED FROM RETEST. BIOS/UEFI checks have already been passed according to the owner. Other evidence also reported Secure Boot enabled; no further BIOS testing is requested.

Acceptance boundary / next proof: Preserve the owner completion declaration and dated record. It is not an independent auditor certification or 21 signed A/B/C gates.

Family	Exact technical scope	A/B/C evidence IDs
39.01	BIOS/UEFI vendor, model and firmware version	KHAHP-39.01A · KHAHP-39.01B · KHAHP-39.01C
39.02	fwupd device/security-attribute inventory	KHAHP-39.02A · KHAHP-39.02B · KHAHP-39.02C
39.03	Secure Boot state and Setup Mode	KHAHP-39.03A · KHAHP-39.03B · KHAHP-39.03C
39.04	TPM presence, version and ownership state	KHAHP-39.04A · KHAHP-39.04B · KHAHP-39.04C
39.05	BIOS/UEFI administrator password manual check	KHAHP-39.05A · KHAHP-39.05B · KHAHP-39.05C
39.06	PXE/CSM/external-boot/manual boot- order policy	KHAHP-39.06A · KHAHP-39.06B · KHAHP-39.06C
39.07	Firmware update, DKMS/MOK and Secure-Boot readiness	KHAHP-39.07A · KHAHP-39.07B · KHAHP-39.07C

40 — Auditd / AIDE / Integrity Monitoring

Observed implementation/evidence: The `dailyaidecheck.timer` was enabled and active. In a SYSMAINT snapshot, audit was enabled but `auditctl -s` showed `pid 0` and `auditctl -l` reported no active rules.

Acceptance boundary / next proof: The owner targets 01–40 completion, but a benign audit event and existing AIDE baseline alert lifecycle were not supplied. Do not claim final Section 40 enforcement acceptance yet.

Family	Exact technical scope	A/B/C evidence IDs
40.01	auditd package/service state	KHAHP-40.01A · KHAHP-40.01B · KHAHP-40.01C
40.02	Audit rule inventory and immutable/finalize policy	KHAHP-40.02A · KHAHP-40.02B · KHAHP-40.02C
40.03	Privileged-command and identity- change audit coverage	KHAHP-40.03A · KHAHP-40.03B · KHAHP-40.03C
40.04	AIDE package/database/configuration state	KHAHP-40.04A · KHAHP-40.04B · KHAHP-40.04C
40.05	File-integrity baseline and scheduled checks	KHAHP-40.05A · KHAHP-40.05B · KHAHP-40.05C
40.06	Audit-log ownership, rotation and tamper resistance	KHAHP-40.06A · KHAHP-40.06B · KHAHP-40.06C
40.07	CIS/DISA accountability-gap analysis	KHAHP-40.07A · KHAHP-40.07B · KHAHP-40.07C

41 — Cron / Timers / Persistence / Supply Chain

Observed implementation/evidence: Scheduled jobs, startup entries and software supply-chain attribution are in the planned final assurance phase.

Acceptance boundary / next proof: Map provenance, signatures and cross-reboot persistence; future work.

Family	Exact technical scope	A/B/C evidence IDs
41.01	cron/anacron job inventory	KHAHP-41.01A · KHAHP-41.01B · KHAHP-41.01C
41.02	systemd timer inventory	KHAHP-41.02A · KHAHP-41.02B · KHAHP-41.02C
41.03	User autostart and desktop persistence inventory	KHAHP-41.03A · KHAHP-41.03B · KHAHP-41.03C
41.04	Package repository and signing-key provenance	KHAHP-41.04A · KHAHP-41.04B · KHAHP-41.04C
41.05	APT pinning/holds and update-policy review	KHAHP-41.05A · KHAHP-41.05B · KHAHP-41.05C
41.06	Unexpected startup scripts/services/modules	KHAHP-41.06A · KHAHP-41.06B · KHAHP-41.06C
41.07	Supply-chain drift and reboot- persistence evidence	KHAHP-41.07A · KHAHP-41.07B · KHAHP-41.07C

42 — Credentials / SSH Keys / Secrets / Backup

Observed implementation/evidence: Credentials and backup planning are documented; a private production-grade isolated restore has not been shown.

Acceptance boundary / next proof: Approve encrypted backups, retention, custody and isolated restoration.

Family	Exact technical scope	A/B/C evidence IDs
42.01	SSH private-key inventory and permissions	KHAHP-42.01A · KHAHP-42.01B · KHAHP-42.01C
42.02	API/token/secret storage locations	KHAHP-42.02A · KHAHP-42.02B · KHAHP-42.02C
42.03	Password-manager selection and vault policy	KHAHP-42.03A · KHAHP-42.03B · KHAHP-42.03C
42.04	FIDO2/YubiKey readiness and hardware-backed keys	KHAHP-42.04A · KHAHP-42.04B · KHAHP-42.04C
42.05	Recovery-code generation and offline storage	KHAHP-42.05A · KHAHP-42.05B · KHAHP-42.05C
42.06	Encrypted backup creation and retention	KHAHP-42.06A · KHAHP-42.06B · KHAHP-42.06C
42.07	Restore test, key recovery and disaster-recovery evidence	KHAHP-42.07A · KHAHP-42.07B · KHAHP-42.07C

43 — CIS / DISA GPOS / KSPP Mapping

Observed implementation/evidence: CIS Debian 13 L2, DISA GPOS/SRG and KSPP applicability references are established.

Acceptance boundary / next proof: Map each applicable requirement to evidence and exceptions; no external certification claim.

Family	Exact technical scope	A/B/C evidence IDs
43.01	CIS Debian 13 Level 2 applicability mapping	KHAHP-43.01A · KHAHP-43.01B · KHAHP-43.01C
43.02	DISA GPOS/SRG objective mapping	KHAHP-43.02A · KHAHP-43.02B · KHAHP-43.02C
43.03	KSPP/kernel-self-protection mapping	KHAHP-43.03A · KHAHP-43.03B · KHAHP-43.03C
43.04	Kicksecure-native equivalent control mapping	KHAHP-43.04A · KHAHP-43.04B · KHAHP-43.04C
43.05	Project-extension control mapping	KHAHP-43.05A · KHAHP-43.05B · KHAHP-43.05C
43.06	Exception/N-A/manual-control rationale	KHAHP-43.06A · KHAHP-43.06B · KHAHP-43.06C
43.07	Evidence citation and no-certification disclaimer	KHAHP-43.07A · KHAHP-43.07B · KHAHP-43.07C

44 — Final Acceptance / Evidence / Scoring

Observed implementation/evidence: A/B/C acceptance methodology is defined for 308 families and 924 gates.

Acceptance boundary / next proof: Final owner/reviewer signoff, negative testing, recoverability and evidence completeness.

Family	Exact technical scope	A/B/C evidence IDs
44.01	Full reboot persistence acceptance	KHAHP-44.01A · KHAHP-44.01B · KHAHP-44.01C
44.02	USER-mode regression acceptance	KHAHP-44.02A · KHAHP-44.02B · KHAHP-44.02C
44.03	SYSMAINT-mode regression acceptance	KHAHP-44.03A · KHAHP-44.03B · KHAHP-44.03C
44.04	Network/DNS/firewall/Tor/USB negative-test acceptance	KHAHP-44.04A · KHAHP-44.04B · KHAHP-44.04C
44.05	Recovery/rollback/backup acceptance	KHAHP-44.05A · KHAHP-44.05B · KHAHP-44.05C
44.06	Evidence completeness, hashes and chain-of-custody bundle	KHAHP-44.06A · KHAHP-44.06B · KHAHP-44.06C
44.07	Final scorecard, residual risk and operator sign-off	KHAHP-44.07A · KHAHP-44.07B · KHAHP-44.07C

6. Historical 01–38 action reconciliation — all 63 IDs

These inherited records were part of the 69-item mixed project backlog. Their existence is not evidence of 63 defects. The current working status below takes newer demonstrations into account but **does not create a formal reviewer signature**. Section 39 BIOS is not retested.

ID	Sec.	Engineering issue	Evidence disposition as of 18 Oct 2026
U01-001	7	/tmp and /dev/shm noexec compatibility decision	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-002	7	/boot/efi-before-/boot fstab ordering review	CONFIGURATION FIX VERIFIED (18 OCT); NEXT-BOOT PERSISTENCE NOT YET ATTESTED
U01-003	2	permission-hardener package/ provenance classifier	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-004	5	sync account shell classification	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-005	12	Unverified kernel tokens / CFI-KCFI claims	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-006	13	vesafb module applicability / dry-run review	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-007	14	USER versus SYSMAINT DKMS evidence reconciliation	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-008	8	Physical GRUB edit-denial test	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-009	6	LUKS key recovery and backup restore	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-010	10	Firmware/preboot DMA protection acceptance	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED

ID	Sec.	Engineering issue	Evidence disposition as of 18 Oct 2026
U01-011	7	Post-reboot SYSMAINT persistence evidence	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-012	7	/home, /var/tmp and noexec compatibility review	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-013	14	Signed-module negative test and update regression	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-014	7	/tmp nodev,nosuid hardening — implemented baseline	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-001	15	Strict Firefox AppArmor confinement is not established	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-002	15	AppArmor service policy and denial regression	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-003	15	USER versus SYSMAINT browser-label persistence	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-004	16	Privileged binary provenance/drift review	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-005	16	gst-ptp-helper capability policy/ compatibility	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-006	17	setup-dist status-files 0777 exception monitoring	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-007	17	Sensitive directory/ACL/umask and tmpfs drift	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-008	18	faillock recovery and self-lockout policy	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-009	18	Passphrase/aging policy ownership	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-010	18	PAM USER/SYSMAINT interaction regression	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-011	19	Network SSH masking and listener persistence	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-012	19	Local SSH AF_UNIX socket and client key policy	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-013	20	Role-specific systemd service drift and listening sockets	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-014	20	Classify unexpected/unused services safely	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-015	21	Unbound systemd exposure score needs design decision	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-016	21	Service sandbox capability regression	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-017	22	Core dump/pstore suppression drift	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-018	22	Crash-forensics versus privacy trade-off	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-019	23	Journal retention, permissions and rotation policy	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-020	23	sdwdate time and journal chronology after reboot	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-021	24	Owner-approved panic/recovery policy and escape paths	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-022	24	Emergency shutdown and controlled recovery evidence	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED

ID	Sec.	Engineering issue	Evidence disposition as of 18 Oct 2026
U25-001	26	External USB keyboard possibly allowed	POLICY CORRECTED: EXTERNAL KEYBOARDS ALLOWED; FUNCTIONAL TEST TO RECONCILE
U25-002	26	Composite/USB network rejection and persistence	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-003	25	Swap crypt mapping and shutdown key lifecycle	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-004	25	RAM wipe shutdown execution and hardware limits	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-005	27	Actual Wi-Fi MAC identity	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-006	27	DHCP identity rotation	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-007	28	Alternate IPv6 network profiles	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-008	29	Unbound response to DHCP renew/ network loss	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-009	30	Unbound-owned authenticated DoT	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-010	30	Direct UDP/TCP 53 egress	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-011	30	Active browser DNS/DoH preferences	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-012	31	Effective firewall output chain	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-013	31	Exact loaded-ruleset persistence	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-014	32	Peer-side discovery proof	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-015	33	Tor Browser routing/DNS/fail-close	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-016	34	Secure-time unit chronology	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-017	25	Original implementation provenance	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-018	26	USB boot media usability versus detection	RECOVERY MEDIA USB AUTHORIZED/DETECTED; ACCESSIBILITY/INSTALL MODE SEPARATE
U25-019	28	IPv6 goal mismatch / explicit policy	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-020	29	USER/SYSMAINT service role semantics	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-021	30	DNSSEC positive/negative evidence signoff	DNSSEC POSITIVE/NEGATIVE OBSERVED; FORMAL CLAIM REVIEW OPEN
U25-022	32	Actual scope of local discovery services	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-023	34	Production clock rollback exception	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U35-001	35	Early-boot entropy and random-seed lifecycle	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U35-002	36	Firefox effective DoH, WebRTC and telemetry behavior	FIREFOX SETTINGS VERIFIED; EXTERNAL BROWSER PATH NEGATIVE STILL OPEN
U35-003	37	Application history, thumbnails, metadata and confinement regression	THUMBNAILS AND REBOOT TEST VERIFIED; OTHER APP BEHAVIOR NOT FULLY SIGNED

ID	Sec.	Engineering issue	Evidence disposition as of 18 Oct 2026
U35-004	38	Camera/microphone hotplug and real lock/lid-resume enforcement	CAMERA/MIC USER DENIAL + BT BLOCK + SHORT LOCK PASS; 5-MIN/LID OPEN

Essential reconciliation decisions

1. **Owner device policy prevails:** external keyboards, two approved mice and USB removable/recovery storage are allowed. Older testing notes requesting keyboard rejection are superseded.
2. **Confirmed changes are not identical to confirmed exploit findings.** Track pre-state, root cause, authorization, exact delta, positive and negative post-state before classifying a true fixed weakness.
3. **Section 07 boot ordering:** immediate root verification passed after correction; avoid reapplying the edit. Preserve backup and collect persistence only when a normal owner reboot occurs.
4. **Section 15 Firefox:** the vendor unconfined placeholder remains an explicitly open MAC enhancement, even though Firefox subprocess seccomp filters exist.
5. **Section 40 audit events:** a working AIDE timer does not prove event coverage. The latest SYSMAINT run observed no active audit rules and PID 0; troubleshoot before a full assurance claim.
6. **Section 39 BIOS:** prior owner completion is accepted as the project process decision and excluded from future retesting; it does not forge formal independent sign-off.

7. Commercial image engineering and supply-chain acceptance

Proposed KHAHP Premium Hardened OS target price: **USD $1,250**; intended distributions are physical ISO/USB, KVM/QCOW2, and VirtualBox/OVA. No public downloadable commercial image or certification is asserted. A clean build must never clone the personal development machine, SSH secrets, random seeds or machine identity.

Clean-image common requirements

Planned release ID	Required demonstration	Current program state
KHAHP-REL-COM-01	Reproducible upstream source	PROPOSED · NOT TESTED
KHAHP-REL-COM-02	Package provenance	PROPOSED · NOT TESTED
KHAHP-REL-COM-03	Build isolation	PROPOSED · NOT TESTED
KHAHP-REL-COM-04	Credential exclusion	PROPOSED · NOT TESTED
KHAHP-REL-COM-05	Unique machine identity	PROPOSED · NOT TESTED
KHAHP-REL-COM-06	Unique SSH host keys	PROPOSED · NOT TESTED
KHAHP-REL-COM-07	SSH access policy	PROPOSED · NOT TESTED
KHAHP-REL-COM-08	Privileges and boot-role separation	PROPOSED · NOT TESTED
KHAHP-REL-COM-09	Default firewall	PROPOSED · NOT TESTED
KHAHP-REL-COM-10	DNS resolver protection	PROPOSED · NOT TESTED
KHAHP-REL-COM-11	Network privacy behavior	PROPOSED · NOT TESTED
KHAHP-REL-COM-12	MAC and device privacy	PROPOSED · NOT TESTED
KHAHP-REL-COM-13	Mandatory access controls	PROPOSED · NOT TESTED
KHAHP-REL-COM-14	Kernel hardening	PROPOSED · NOT TESTED
KHAHP-REL-COM-15	Audit visibility	PROPOSED · NOT TESTED
KHAHP-REL-COM-16	File integrity monitoring	PROPOSED · NOT TESTED
KHAHP-REL-COM-17	Upgrade safety	PROPOSED · NOT TESTED
KHAHP-REL-COM-18	Backup and restoration	PROPOSED · NOT TESTED
KHAHP-REL-COM-19	Release signing	PROPOSED · NOT TESTED
KHAHP-REL-COM-20	Customer documentation and licensing	PROPOSED · NOT TESTED

Physical ISO / USB requirements

Planned release ID	Required demonstration	Current program state
KHAHP-REL-ISO-01	UEFI USB boot	PROPOSED · NOT TESTED
KHAHP-REL-ISO-02	Installer storage detection	PROPOSED · NOT TESTED
KHAHP-REL-ISO-03	Fresh encrypted installation	PROPOSED · NOT TESTED
KHAHP-REL-ISO-04	Encryption recovery	PROPOSED · NOT TESTED
KHAHP-REL-ISO-05	Secure Boot enrollment	PROPOSED · NOT TESTED
KHAHP-REL-ISO-06	Firmware Setup Mode	PROPOSED · NOT TESTED
KHAHP-REL-ISO-07	TPM optional path	PROPOSED · NOT TESTED
KHAHP-REL-ISO-08	Intel and AMD support	PROPOSED · NOT TESTED
KHAHP-REL-ISO-09	Graphics and desktop	PROPOSED · NOT TESTED
KHAHP-REL-ISO-10	Wireless networking	PROPOSED · NOT TESTED
KHAHP-REL-ISO-11	Ethernet and hotplug	PROPOSED · NOT TESTED
KHAHP-REL-ISO-12	USBGuard defaults	PROPOSED · NOT TESTED
KHAHP-REL-ISO-13	Installer device safety	PROPOSED · NOT TESTED
KHAHP-REL-ISO-14	DMA/IOMMU capabilities	PROPOSED · NOT TESTED
KHAHP-REL-ISO-15	Signed module lifecycle	PROPOSED · NOT TESTED
KHAHP-REL-ISO-16	Physical webcam/microphone	PROPOSED · NOT TESTED
KHAHP-REL-ISO-17	Power and RAM-wipe process	PROPOSED · NOT TESTED
KHAHP-REL-ISO-18	BIOS recovery restrictions	PROPOSED · NOT TESTED
KHAHP-REL-ISO-19	Multi-boot/firmware resilience	PROPOSED · NOT TESTED
KHAHP-REL-ISO-20	Fresh-install two-device comparison	PROPOSED · NOT TESTED

KVM / QCOW2 requirements

Planned release ID	Required demonstration	Current program state
KHAHP-REL-KVM-01	QCOW2 import and boot	PROPOSED · NOT TESTED
KHAHP-REL-KVM-02	Fresh clone identity	PROPOSED · NOT TESTED
KHAHP-REL-KVM-03	Virtual firmware choice	PROPOSED · NOT TESTED
KHAHP-REL-KVM-04	Virtual TPM support	PROPOSED · NOT TESTED
KHAHP-REL-KVM-05	Host-guest firewall isolation	PROPOSED · NOT TESTED
KHAHP-REL-KVM-06	Guest disk encryption	PROPOSED · NOT TESTED
KHAHP-REL-KVM-07	Virtio device support	PROPOSED · NOT TESTED
KHAHP-REL-KVM-08	KVM snapshot and restore	PROPOSED · NOT TESTED
KHAHP-REL-KVM-09	KVM version regression	PROPOSED · NOT TESTED
KHAHP-REL-KVM-10	QCOW2 sanitized release	PROPOSED · NOT TESTED

VirtualBox / OVA requirements

Planned release ID	Required demonstration	Current program state
KHAHP-REL-OVA-01	OVA packaging and import	PROPOSED · NOT TESTED
KHAHP-REL-OVA-02	Unique identity on import	PROPOSED · NOT TESTED
KHAHP-REL-OVA-03	Clipboard and drag-drop	PROPOSED · NOT TESTED
KHAHP-REL-OVA-04	Shared-folder isolation	PROPOSED · NOT TESTED
KHAHP-REL-OVA-05	USB passthrough restriction	PROPOSED · NOT TESTED
KHAHP-REL-OVA-06	VirtualBox networking	PROPOSED · NOT TESTED
KHAHP-REL-OVA-07	Guest addition safety	PROPOSED · NOT TESTED
KHAHP-REL-OVA-08	Virtual disk recovery	PROPOSED · NOT TESTED
KHAHP-REL-OVA-09	Supported host matrix	PROPOSED · NOT TESTED
KHAHP-REL-OVA-10	OVA clean release gate	PROPOSED · NOT TESTED

8. Planned completion of Sections 41–44

Section	Required work	Release proof
41 — Persistence / supply chain	Cron/timer/autostart classification, signing and package origin, two-boot drift	Attributed, bounded and stable scheduled activity
42 — Credentials / backup / recovery	Secret inventory without disclosure; owner-approved encrypted backups; isolated restore	Successful recovery and sample hash comparison; custody controls
43 — CIS/DISA/KSPP applicability	Map every relevant requirement, native equivalent and justified N/A	Evidence-linked applicability map; no official certification claim
44 — Final A/B/C sign-off	Independent review, complete negative/ regression and rollback proof, dated acceptance	308 families × A/B/C accepted with documented risk exceptions

9. Compliance, controlled change and public disclosure

Execution roles: USER runs normal browser and read-only checks without sudo. SYSMAINT is a separate privileged boot, with independently verified source digests and a root-owned copy for execution. Privileged scripts are never executed in place from a USER-writable transport path.

Change gate: evidence → verified root cause → minimal diff → backup/recovery → owner approval → apply → immediate postflight → bounded negative test → relevant reboot/update persistence → dated A/B/C acceptance.

Publication safety: No hostnames, private identifiers, IP/MAC addresses, raw firewall and browser history, SSH material, real encryption IDs/keys, authentication secrets or raw confidential reports. This register publishes redacted outcomes; underlying logs remain in private custody. Historical Section 37 self-hashing manifest errors must be identified rather than presenting those archives as fully hash-valid.

Reference frameworks: CIS Debian 13 Level 2 Workstation, DISA GPOS SRG objectives, KSPP, Debian and Kicksecure security practices. These are engineering references rather than attestation of compliance or certification.

Public challenge submissions: provide machine-independent provenance, threat model, safely reproducible tests, intended/observed outcomes, minimized changes, rollback proof, and limitations. Do not perform unauthorized external testing or disclose user-specific security details.

10. Evidence source ledger for this revision

- Pre-existing public register v3.0 (17 October 2026), 308 technical family labels and 60 planned release tests.
- Private KHAHP master v2.0 and 924-gate proof-claims CSV dated 18 October 2026 (all historical signatures OPEN).

- KHAHP compliance standard v3.0, field manual v2.0 and 63-item historical 01–38 closeout list.
- USER 01–38 read-only observations of 18 October 2026; SYSMAINT 65-check archive with validated hashes and zero changes.
- Later USER direct camera/mic open-denial, USBGuard inventory, Firefox process-label/prefs, DNSSEC, AIDE timer and manual short-lock observations.
- Operator-shared SYSMAINT terminal evidence of `/etc/fstab` boot-order correction and clean `findmnt --verify --verbose` validation.
- Section 39 BIOS status: owner declaration, no new test requested.

Evidence limitation: No complete Section 40 audit-event/AIDE alert lifecycle report or formally signed 924-gate acceptance register was supplied in this release; unsupported completion claims were intentionally not synthesized.